

Deterministic Optimization

Nonlinear Optimization
Modeling – Approximation and
Fitting

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Linear Equations, Norm, and Least
Squares Problem

Nonlinear Optimization Modeling

Learning Objectives for this module

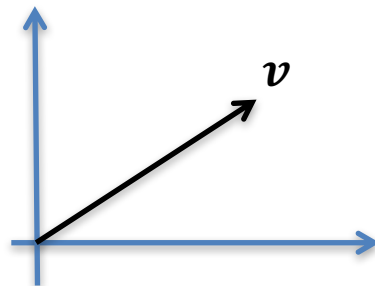
- Explore nonlinear optimization models with applications in approximation and function fitting

A Simple Question

- **Nonlinear** optimization involves nonlinear objective and/or nonlinear constraints
- The simplest nonlinear optimization model comes from the following question:
 - “Given a matrix $A \in R^{m \times n}$ and a vector $b \in R^m$, where $m > n$, how to find a solution x that satisfies $Ax = b$ as closely as possible?”
 - Why “as closely as possible”?
 - Because there are more equations than unknowns ($m > n$), not all equations may be satisfied simultaneously by a solution x .
 - How to model “as closely as possible”?
 - Need to measure how big the error $(Ax - b)$ is.
 - The error $(Ax - b)$ is a vector, so we need to measure vectors.

Norm: Measure the Length of a Vector

- We need to measure the “length” of a vector
- Mathematically, the “length” is called the **norm** of a vector
- For instance, the usual way of defining the length of a vector:
 - $v = (x_1, x_2)$, length of v is $\sqrt{x_1^2 + x_2^2}$
 - This is called the **l_2 -norm** of v , $\|v\|_2 = \sqrt{x_1^2 + x_2^2}$
 - Also called **Euclidean norm**
 - In higher dimension, $v = (x_1, \dots, x_n)$
 - $\|v\|_2 = \sqrt{x_1^2 + \dots + x_n^2}$



Least Square Problem

- The error term is given as $Ax - b = \begin{bmatrix} a_1^\top x - b_1 \\ \vdots \\ a_m^\top x - b_m \end{bmatrix}$
- If there exists a solution x such that $Ax = b$, then the error vector $(Ax - b)$ is zero, therefore, the norm of the error is 0.
- If there does not exist a solution x such that $Ax = b$, then can we find a solution x that makes the norm of the error as small as possible?

- $\min_x ||Ax - b||_2 = \min_x \sqrt{\sum_{i=1}^m (a_i^\top x - b_i)^2}$
- Equivalently, $\min_x ||Ax - b||^2 = \min_x \sum_{i=1}^m (a_i^\top x - b_i)^2$

Summary

- We learned:
 - How to measure the “length” of a vector by its Euclidean norm
 - How to formulate a least squares problem